STA2002: Probability and Statistics II
Midterm Exam

March 24, 2022

The questionnaire comes with three questions. You need to answer them all. The number of points each question carries is stated next to it. The total is 100. You have 80 minutes in order to do so. You are entitled to bring along to the exam class two A4 sheets of papers which you have prepared in advance. Finally, you are asked to present a valid Student I.D. Card for inspection during the examination session.

1. (25) Let $X_1, X_2, \ldots, X_{12}$ be a series with 12 entries. It is known that $X_3 = 4$, $X_5 = 6$, $\overline{X} = 5$ and that $\text{Var}(X) = 4$. Let $Y_1, \ldots, Y_{12}$ be a similar series but with two exceptions $Y_3 = 3$ and $Y_5 = 7$.

   (a) (5) what are the standardization of X_3 and X_5?

   (b) (10) what is the value $\overline{Y}$?

   (c) (10) what is the value $\text{Var}(Y)$?

   Solution:

   (a) -0.5 and $+0.5$, respectively.

   (b) the mean has not changed: $\overline{Y} = 5$

   (c) in the sum of squares leading to the variance, two entries of value of 1, were replaced by 4. The other ten have not changed. This leads to a total increase of 6 in this summation, adding a value of $6/12$ to the variance, making the new variance equal to $4 + 0.5 = 4.5$.

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2. (30) A sampled of some size n was conducted on three variables, X, W and Y. Both X and W are measured in kilograms, while Y is measured in meters. The following figures were computed $\overline{X} = 4.5$, $\overline{W} = 3.2$ and $\overline{Y} = 11.2$, $\text{Var}(X) = 4.05$, $\text{Var}(W) = 4.36$ and $\text{Var}(Y) = 22.56$. Finally, $\text{Cov}(X, Y) = 8.4$ and $\text{Cov}(W, Y) = 6.76$.

   (a) (6) compute $\text{Corr}(X, Y)$ and $\text{Corr}(W, Y)$.

   (b) (6) what is the regression line of Y on X. Next to the numerical values for the slope and for the intercept state their units of measurement.

   (c) (6) repeat the previous item but now replace X with W.

   (d) (6) which of the above two regression lines do you recommend to use?

   (e) (6) for the regression line you have stated in the previous item, compute the variance of Y it explains. How much is it in percents?

Solution:

(a) $\text{Corr}(X, Y) = \frac{\text{Cov}(X,Y)}{\text{SD}(X)\text{SD}(Y)} = \frac{8.4}{\sqrt{4.05 \times 22.56}} = 0.8788$
$\text{Corr}(W, Y) = \frac{\text{Cov}(W,Y)}{\text{SD}(W)\text{SD}(Y)} = \frac{6.76}{\sqrt{4.36 \times 22.56}} = 0.6816$

(b) The regression line is $\frac{y-\bar{Y}}{\text{SD}(Y)} = \text{Corr}(X, Y)\frac{x-\bar{X}}{\text{SD}(X)}$. In particular,

$$\frac{y - 11.2}{\sqrt{22.56}} = 0.8788 \times \frac{x - 4.5}{\sqrt{4.05}}$$

$$\Rightarrow y = 2.0741x + 1.8667$$

The slope 2.0741 is measured in meters per kilogram, while in intercept 1.8667 is measured in meters.

(c) $y = 1.5505x + 6.2385$
The slope 1.5505 is measured in meters per kilogram, while in intercept 6.2385 is measured in meters.

(d) The regression line of Y on X: $y = 2.0741x + 1.8667$.

(e) $\sum_{i=1}^{n}(\hat{Y}_i - \bar{Y})^2 = \text{Corr}^2(X, Y)n\text{Var}(Y)$. In percent, it is $\text{Corr}^2(X, Y) = 77.23\%$.

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3. (45) Let $X \sim \chi^2_{(2m)}$ for some integer m.

   (a) (4) what is the density function of X?

   (b) (8) what are the first two moments of X?

   (c) (4) what is the variance of X?

   (d) (10) based on a random sample X_i, $1 \leq i \leq n$, all of which are independent and distributed as X, design by the method of moments, two estimators for m based on the two moments derived above. Note that they need not be integers.

   (e) (6) what is the likelihood function of the random sample?

   (f) (9) what is the MLE for m and how it is related to the sample geometric mean? Note that it needs to be an integer. Hint: Consider the ratio between the likelihood functions for two consecutive values for m.

   (g) (4) What is the MLE for 2m?

   Solution:

   (a) X is fact follows a Gamma distribution with parameters $2m/2 = m$ and $1/2$. Hence,

$$f_X(x) = \frac{(1/2)^m}{(m-1)!} x^{m-1} e^{-x/2}, \ x \geq 0.$$

   (b) We know that if $X \sim \Gamma(\alpha, \beta)$ then $\mathrm{E}(X) = \alpha/\beta$, $\mathrm{E}(X^2) = \alpha(\alpha+1)/\beta$ and $\mathrm{Var}(X) = \alpha/\beta^2$. Hence, $\mathrm{E}(X) = m/(1/2) = 2m$, $\mathrm{E}(X^2) = m(m+1)/(1/2)^2 = 4m(m+1)$

   (c) $\mathrm{Var}(X) = m/(1/2)^2 = 4m$.

   (d) Since $\mathrm{E}(\overline{X}) = 2m$, the first estimator is derived from $\overline{X} = 2\hat{m}$. Hence,

$$\hat{m} = \frac{\overline{X}}{2}.$$

   Since $\mathrm{E}(\overline{X^2}) = 4m(m+1)$, the second estimator is derived from $\overline{X^2} = 4\hat{m}(\hat{m}+1)$. This leads to a quadratic equation in $\hat{m}$. Hence,

$$\hat{m} = \frac{-1 + \sqrt{1 + \overline{X^2}}}{2}.$$

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Note that the other root leads to a negative value.

(e)

$$L(X_1, \ldots, X_n; m) = \Pi_{i=1}^n \frac{(1/2)^m}{(m-1)!} X_i^{m-1} e^{-X_i/2}$$

$$\frac{(1/2)^{nm}}{((m-1)!)^n} (\Pi_{i=1}^n X_i)^{m-1} e^{-\sum_{i=1}^n X_i/2}$$

(f)

$$L(X_1, \ldots, X_n; m+1) = \frac{(1/2)^{n(m+1)}}{(m!)^n} (\Pi_{i=1}^n X_i)^m e^{-\sum_{i=1}^n X_i/2}$$

From the above we learn that

$$\frac{L(X_1, \ldots, X_n; m+1)}{L(X_1, \ldots, X_n; m)} = (\frac{1}{2m})^n \Pi_{i=1}^n X_i.$$

Since $X \geq 0$, this ratio is clearly monotone decreasing with m and we look for the first m such that

$$(\frac{1}{2m})^n \Pi_{i=1}^n X_i \leq 1$$

or

$$\frac{(\Pi_{i=1}^n X_i)^{\frac{1}{n}}}{2} \leq m$$

Now $\hat{m}$ needs to be an integer, so it is the ceiling of what is written in the left hand side, which in fact is the ceiling of half the geometric sample mean.

(g) The MLE for $2m$ is $2\hat{m}$.

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